

SAFETY RULES IN ICT LABORATORIES - 2025

A wide variety of testing equipment is available for your research at ICT. Students are required to perform various activities that could be dangerous. This document is a guide to ensure the safe and effective work of all students at ICT.

Listed below are the general safety guidelines that apply to all activities in the laboratory. In addition to these rules, more specific guidelines are associated with each piece of equipment. Equipment-specific rules will be introduced during individual training/verification process. Lab space includes but is not limited to the labs at ICT, MPF, the shed and the ATLAS yard.

General Safety Rules

1. Complete the Division of Research Safety online "general lab safety" program and provide the certificate of completion to the Research Engineers. <http://research.illinois.edu/> : "Training" -> "Online Researcher Training Portal" -> login using UIUC credentials -> "Training Library" -> search for " laboratory safety training " -> click on the "Laboratory Safety Training" box.
2. Read and understand the UIUC Laboratory Safety Guide. Online at <https://www.drs.illinois.edu/site-documents/LaboratorySafetyGuide.pdf> , or obtain a printed copy from the Research Engineers.
3. NEVER perform work that you are uncomfortable performing. (See also #12)
4. Each researcher that desires to work in the lab must complete training on general safety at the ICT facility. All undergraduate and graduate students must sign a compliance form (Appendix D) acknowledging that they have received this training. This must be completed prior to performing any work at ICT.
5. Normal lab hours are from 8am to 5pm Monday through Friday. No person should work alone in labs or office space outside of normal lab hours. When the building is occupied outside of normal lab hours, there should be no fewer than two people in the building. If you wish to work after hours, it is your responsibility to find a second person to accompany you. This person must also be faculty, staff, or student of ICT that has undergone the safety training.
6. All students working in the lab must be dressed appropriately to the nature of the work being performed; see Appendix A.
7. Any electrical extension cords or power strips must follow rules listed in Appendix B.
8. First aid kit must be placed in the lab. Anyone working in the lab must know the location of the first aid kit. If anything from the first aid kit is used the incident must be reported to a lab supervisor or research engineer so the kit can be resupplied. Using first aid kit supplies is encouraged in the case of an injury.

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9. An oil spill kit will be stored in the Solvent room. Any oil spilled on the floor must be cleaned up immediately.
10. A fire extinguisher must be readily available. Anyone working in the lab is responsible for knowing the location of the fire extinguisher. If a fire extinguisher is used for any reason, the incident must be reported to a lab supervisor or research engineer so the extinguisher can be recharged.
11. No horseplay will be permitted in the lab.
12. No headphones or ear buds are allowed in the lab areas. Audible music may be played in the lab or work space if all those in the area are in agreement and the volume of the music does not interfere with normal communication and with safe operation of equipment.
13. The lab area must be kept clean. It is impossible to maintain safe and high quality work in a messy environment. Users are responsible to keep their work area organized and clean. If the lab is not kept clean, lab privileges will be taken away.
14. Practice common sense. If your instinct tells you not to perform an action, don't do it. Likewise, if you are not completely confident when using a machine, don't use it. Most lab accidents are a result of lack of knowledge, not carelessness. If you have any doubt, consult the Research Engineers.
15. The alarm for the main ICT building will be disarmed at 8am and armed at 5pm Monday through Friday. MPF may remain armed if no one has used it in a given day. Assume MPF is armed until you verify the alarm status. If an individual comes to the lab outside of these hours, they are responsible for disarming and re-arming the alarms.
16. If an alarm is not armed upon leaving a building after-hours or is triggered by a student and Police are called to the ICT campus, penalties will be enforced.
17. Appendix C provides executive regulations.
18. Any observed violation of the above rules will result in suspension from the lab. The duration of the suspension will be based on the nature of the violation and can range from one day to a permanent ban. Research Engineers, or the Director of ICT, will determine the appropriate penalty.

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Appendix A

Personal Protective Equipment (PPE) Requirements for Lab Areas:

1. Safety-toe shoes and long pants made from sturdy material are required. A user of the lab areas should not wear loose clothing. Long hair must be tied back at all times. Long sleeved shirts are recommended at all times. No sweatpants, yoga pants, pajama pants or shorts will be allowed.
2. Eye safety glasses will be worn in the lab and testing areas at all times. Normal corrective eyewear is sufficient for general lab activities. However, if engaged in an impact type work, corrective eyeglasses must meet ANSI Z87+ standards and have additional side shields. Otherwise, additional safety glasses must be worn over the corrective glasses.
3. Any time a heat source and solvents or flammable materials are being used, a flame resistant lab coat and a full face shield must be worn.
4. Neckties, scarves, necklaces, bracelets and other loose articles must be removed prior to working in the lab.
5. Hard hats must be worn at all times when lifting heavy objects with the crane.
6. If you have any doubt, the Research Engineers will decide on the appropriate items for wear.
7. If any individual is found without the proper PPE for the task they are performing, they will be penalized as previously mentioned in General Safety Rules Item No. 17.

Appendix B

Safety Rules While Using Electrical Devices:

1. Extension cords shall contain the number of conductors and gage of wire required for the service plus an equipment ground wire.
2. Extension cords shall be protected from damage (including that caused by foot traffic, vehicles, sharp corners, protections, and pinching).
3. Flexible cord shall be used only in continuous lengths without splice or tape. Patched, oil-soaked, worn, or frayed electric cords or cables shall not be used.
4. Plugs and receptacles shall be kept out of water unless they are of an approved submersible type.
5. All equipment used in damp areas (and preferably everywhere) should be connected to a Ground-Fault Circuit Interrupting (GFCI) device. If the GFCI device is not integral to the wall receptacle, a portable GFCI breaker should be used as close to the wall receptacle as practicable.
6. No extension cord should be used permanently in the lab. If a piece of equipment requires power and its cord will not reach an outlet, a new outlet will need to be installed.

Appendix C

Executive Policy on Implementation and Execution the Safety Rules:

1. Director of the lab is responsible for implementation and execution of safety rules in ICT. He or she may designate a person; e.g. lab coordinator/manager or research engineer who will provide training and execute safety rules on daily basis.
2. Principal Investigator (PI) and lab coordinator/manager are responsible for organization, cleanliness, and safety in testing areas. Safety rules for the designated sections must comply with the General Safety Rules listed above.
3. Students and staff working on the experimental research project are allowed in the designated sections, only. Visitors are allowed in the lab under the condition that they follow general rules established for the lab and specific rules for designated section. All visitors must wear safety glasses while in labs, and other PPE as appropriate for the situation. All visitors to lab areas must be accompanied by an ICT research engineer or a person designated by them.
4. Each new student will be introduced to general and specific safety rules prior to initiation meeting associated with new project (he/she will receive a copy of this document via e-mail to gain familiarity with the rules). He/she will be verified during an initiation meeting and asked to sign a compliance form; Appendix D.
5. Faculty and staff of ICT will pay attention to students and staff working in the lab. Faculty and staff shall react to any obvious violations of the safety rules.
6. Everyone is responsible for the observance and compliance of the safety rules. Anyone (students especially) can and should take action to make sure that each and every person operates in a safe manner.

Appendix D

**Civil and Environmental Engineering ICT
Laboratory Safety Rules**

COMPLIANCE FORM

I have read, understood, and will comply with the rules outlined in the Civil and Environmental Engineering ICT Laboratory Safety Rules. I will take full responsibility for any action that may happen while using the ICT Laboratories.

STUDENT:

Last Name (print)

First Name (print)

Signature

Date Signed

UIUC Email address

PRINCIPAL INVESTIGATOR:

Last Name (print)

First Name (print)